

Ethan Wheeler

Student Engineer and Researcher

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EDUCATION

University of Tennessee Knoxville

B.S. Biomedical Engineering, Neuroscience minor

Knoxville, TN

Expected May 2027

Roane State Community College

Associate of Science, General Studies

Oak Ridge, TN

May 2020

RELEVANT COURSEWORK

Mechatronics | Signal Analysis | System Dynamics | Material Science | Circuits | Neuroscience Lab Methods

SKILLS

Technical: Fusion 360, Blender, Adobe Suite, DaVinci Resolve, Teensy/Arduino firmware, CAN bus, brushless FOC motor control, IMU sensors, drone piloting (FAA Part 107 certified)

Professional: International research experience, technical writing, oral/visual presentation, laboratory procedures, field work

RELEVANT PROJECT EXPERIENCE

Wearable Robotic Third Arm

May 2026 – Present

Designed and built a wearable 5-axis robotic arm that copies the movements of my arm using IMU trackers I built. It runs five brushless motors over a CAN bus and has a tendon-driven 3-finger gripper. Wrote the control software with AI assistance and tested it on the hardware. Presenting it live at the UTK engineering expo, September 29, 2026.

Resumé Arm

October 2025 – February 2026

Self-designed machine with a robotic arm and paper dispenser that hands out copies of my resume

- Researched capabilities of hardware I had on hand and what I would need to design the machine
- Sourced parts and designed the mechanisms and interfaces in CAD, rapidly iterated and 3D printed them
- Built a frame and tested extensively while wiring and coding the arm and dispenser
- Adjusted the design during building to correct unforeseen issues
- Optimized for cost as it was self-funded, total cost \$600

Building Electric Bikes

October 2024 – May 2025

Mountain and road bike conversion to electric

- Researched how current products work and what it would take to make my own battery packs and convert my bikes
- Bought cost effective hardware and kept to a budget of \$250
- Learned how to make battery packs safely and used them to power my bikes for many months

RELEVANT WORK EXPERIENCE

U.S. Public Health Service – Indian Health Service

Portland, OR

Engineer Officer, Junior COSTEP (Ensign)

June 2025 – August 2025

- Worked on the Portland Area's largest design-build contract for medical facilities, including a school based health center
- Reviewed mechanical, civil, electrical, and biomedical systems, made cost estimates, and monitored active projects
- Inspected water treatment systems on Indian reservations, including pump systems, storage tanks, and treatment systems
- Performed environmental audits for the EPA for reservation sites
- Presented my summer work at the national JRCOSTEP symposium

Karolinska Institute

Stockholm, Sweden

Research Assistant – Supervisors: Nick Mitsios and Jan Mulder

January 2024 – May 2024

- Performed immunochemistry fluorescent staining on human brain tissue to find where the protein NEUROD6 is expressed
- Prepared samples, did antigen retrieval, and ran the antibody incubations and TSA staining
- Analyzed the scanned images to see which cell types expressed the protein in the cortex, hippocampus, and cerebellum
- Found the protein mostly in neurons, including about 80% of cerebellar neurons
- Presented the findings on a research poster at the end of the term

Kelvion

Knoxville, TN

Quality Engineer Intern

May 2024 – August 2024

- Worked as a quality tech inspecting heat exchanger manufacturing and shadowed engineers
- Helped with reading engineering drawings and tracking supply problems

University of Tennessee NIMBioS Lab

Knoxville, TN

Drone Pilot

June 2023 – May 2025

- Piloted a Mavic 2 Enterprise drone to record thermal and visual imagery for a spatial analysis lab
- Registered the aircraft and completed airworthiness checks before each flight

Freelance Photography and Animation

Tennessee

Self-employed – contract photography and animation work while traveling

2020 – 2023